Magnetic field acceleration of catalyzed para- to orthohydrogen conversion for cooling superconducting motors

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Endothermic para-orthohydrogen conversion has the potential to cool compact light-weight superconducting motors to support high fuel efficiency aviation. Externally applied magnetic fields from the superconducting motor can enhance catalyst activity to deliver larger cooling loads with smaller parahydrogen catalytic converter sizes. This study experimentally investigated the influence of an externally applied magnetic field on the para-orthohydrogen conversion rate on a commercial ferric oxide catalyst (Ionex type OP) at temperatures relevant to superconductor motor cooling. Experiments indicated that a 2 kOe applied magnetic field accelerated the catalysis rate by 46 % at a temperature of 40 K, whereas 10 % magnetic deceleration of the catalysis rate was observed at a temperature of 90 K. Overall, utilising the magneto-catalytic synergy could reduce the required catalyst volume for cooling superconducting motors. Kinetic analysis and catalyst magnetic property characterisation indicate that the applied magnetic field reduced the catalyst magnetic inhomogeneity and the energy barrier for conversion.

A	Pre-exponential factor, Bara/min
C	Molar concentration, mol/L
E	First order activation energy, J/mol
F	Molar flow rate, mol/min
H_O	Orthohydrogen enthalpy, kJ/kg
H_P	Parahydrogen enthalpy, kJ/kg
$\Delta H_{PO+sensible}$	Enthalpy change due to change Po conversion and sensible heat transfer
$\Delta H_{sensible}$	Enthalpy change due to sensible heat transfer only
k	First order rate constant, 1/min
$k_{baseline}$	first order rate constant with no applied magnetic field, 1/min
k_{magnet}	First order rate constant with applied magnetic field, 1/min
$\dot{m}$	Hydrogen mass flow rate, kg/s
n	Exponent of pressure in rate constant expression, -
P	Pressure, Bara
Q	Cooling duty, kW
R	Ideal gas constant, J/mol/K
t	Residence time in catalytic converter, min
T	Temperature, K
V	Catalyst bed total volume inclusive of void fraction, L
$\dot{v}$	Actual hydrogen volumetric flow rate L/min
X	p-H_2 fraction, -
X_1	Inlet p-H_2 fraction, -
X_2	Outlet p-H_2 fraction, -
X_e	Equilibrium p-H_2 fraction, -
X_{exp}	Experimental outlet p-H_2 fraction, -
X_{model}	Outlet p-H_2 fraction predicted by kinetic model, -
ε	Extent of p-H_2 conversion, -

1. Introduction

The emergent technologies of liquid hydrogen (LH_2) electric propulsion and superconducting motors present the opportunity for high mass efficiency propulsion. Pairing these technologies could offer a synergy where regasification of the LH_2 fuel could provide cooling to the superconducting motor, streamlining the aircraft energy management. Without this synergy, additional subsystems for regasification of the LH_2 and cryogenic refrigeration of the superconducting motor would be required, both drawing from the mass and energy budget of the aircraft.

Superconducting (SC) motors using MgB_2 and REBCO wire typically require cooling to temperatures between 20 and 90 K [1]. The high current density and low electrical resistance of these motors have been demonstrated to achieve 50 % reductions in the mass and volume of the motor compared to a non-superconducting equivalent [2] and electrical efficiencies over 97% [3, 4]. SC motors can result in high specific power values over 5 kW/kg [5, 6] while conventional induction motors typically achieve a specific power of 1 kW/kg [6]. The higher efficiency and mass and volume savings of SC motors can result in an approximate 10% increased fuel efficiency [7-9].

SC motors are currently being explored for aviation applications where a specific power over 25 kW/kg is targeted [10, 11]. This could be achieved using LH_2 as the fuel and coolant, as cryocooler based cooling increases the mass and therefore reduces the specific power. In a conceptual aircraft design using 3 MW power output SC motors, using a cryocooler would reduce the specific power from 22 kW/kg to 12 kW/kg [9]. Emergent aircraft SC motor designs are exploring the use of re-gasified LH_2 for the coolant such as the 2 MW power output motor developed by Airbus [12]. Assessing the suitability of LH_2 SC motor cooling requires understanding the amount of cooling deliverable and the required mass and volume required to do so.

Cooling of SC motors can be provided by both the sensible heat and endothermic parahydrogen ($p\text{-}H_2$) to orthohydrogen ($o\text{-}H_2$) spin isomer conversion (henceforth referred to as 'P→O') that is offered by the regasification of LH_2. The extent of cooling delivered by P→O conversion is a function of the equilibrium $p\text{-}H_2$ fraction and the enthalpy change of the hydrogen isomers both of which are functions of temperature. Maximising the available cooling power for SC motors requires maximising the P→O conversion rate through the use of magnetic catalysts. A ferric oxide catalyst powder such as the "Ionex type OP" catalyst produced by molecular Products [13] (referred to as "Ionex" henceforth) is typically utilised for this based on its balance of activity, production scale and cost. A general window of effective P→O cooling exists between temperatures of 40 and 110 K based on the LH_2 being stored at its equilibrium composition of nearly 99.8 % $p\text{-}H_2$. This is shown in Figure 1 where the P→O cooling duty is compared to sensible cooling alone through enthalpy balances. A P→O cooling contribution maximum exists as conversion at T = 40 K only represents an approximate 0.1 $p\text{-}H_2$ fraction change while the approximately 700 kJ/kg enthalpy of P→O conversion begins to reduce above T = 90 K leading to less cooling per mass of catalyst. Inducing P→O conversion in the regasification of $p\text{-}H_2$ at T = 90 K could increase the amount of cooling provided by 50 % compared to the sensible heat alone [14].

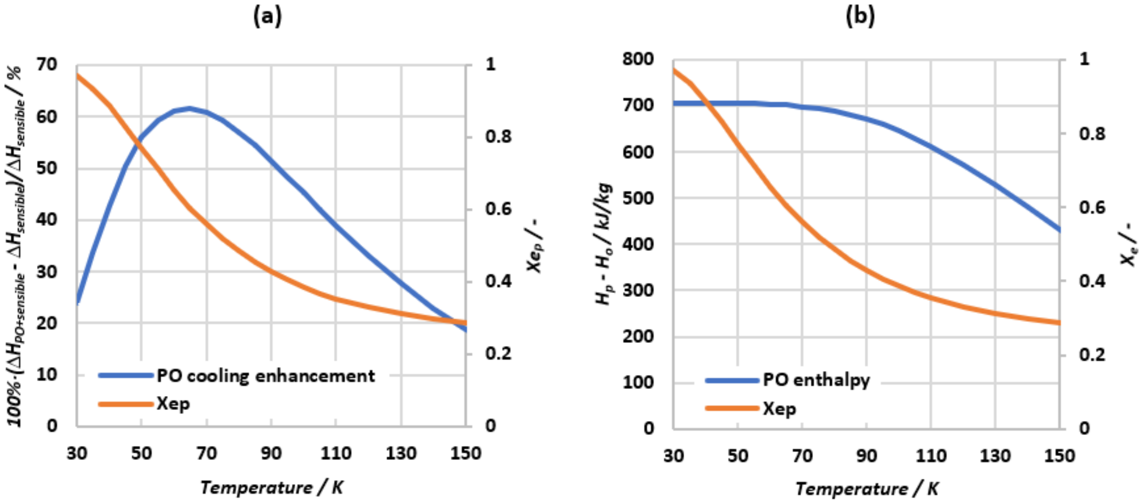


Figure 1: P→O conversion cooling with respect to temperature using equations of state of [15] at 2 Bara. (a) P→O cooling enhancement compared to sensible cooling with p-H_2 only and contrasted to the equilibrium p-H_2 fraction. (b) Enthalpy of P→O conversion and equilibrium p-H_2 fraction from the statistical thermodynamics of [16].

The typical 0.7 – 1 Tesla magnetic field produced by SC motors [4, 7, 11] presents the opportunity to leverage the magneto-catalytic effect to enhance P→O catalyst activity. This potential synergy could achieve higher cooling to mass and volume ratios but has not been publicly published on. This study experimentally measured the P→O conversion rate over the conventional Ionex catalyst with and without an applied magnetic field at T = 45 – 90 K to progress the technology readiness of P→O coolers. The kinetic results were then applied to evaluate the high-level value proposition of P→O cooling for aircraft SC motors. Insights on the magneto-catalytic effect could help develop theories on the P→O conversion mechanism to support the development of higher activity catalysts

2. Background

Externally applied magnetic fields have been observed to accelerate the P→O conversion rate over transition metal oxides catalysts such as Cr_2O_3 [17] and suggest that a magneto-catalytic effect may also be present for the commercial Ionex ferric oxide catalyst. The magneto-catalytic effect is linked to the fundamental P→O conversion catalysis mechanism which is briefly presented to support discussion on the influencing factors on the magneto-catalytic effect reported in literature and this work. Numerous reviews of the magneto-catalytic effect have been published [18-20] addressing the phenomena from a physics perspective. The intent of this review is to explore the influencing factors on the magneto-catalytic effect from an applied engineering perspective.

Conversion mechanism

Orthohydrogen (o-H_2) refers to H_2 molecules with symmetric nuclear spins, whereas parahydrogen (p-H_2) refers to H_2 molecules with anti-symmetric nuclear spins. The nuclear spin represents the angular moment of the proton which results in each proton's magnetic dipole moment. Angular momentum must be conserved during P→O conversion and means that a single p-H_2 molecule cannot spontaneously convert to o-H_2 without an externally applied angular moment [21]. Angular momentum is transferred through contact with a heterogenous catalyst. This process is non-dissociative in nature below T = 100 K where an applied inhomogeneous magnetic moment causes the nuclear spin of one of the protons to flip. This causes the H_2 rotational energy level to change via the Pauli exclusion principle, resulting in an endothermic process for P→O conversion. An inhomogeneous magnetic field on the length scale of a H_2 molecule is required to create a magnetic torque [20] for spin state change as a homogenous magnetic field applied to H_2 molecules will not induce conversion. Recent studies have linked an inhomogeneous electric potential of a solid catalyst as a means of spin state change [22] and is likely related to the angular momentum conservation through electromagnetism.

Achieving the required magnetic torque involves close contact of the H_2 molecule with the applied magnetic torque through van der Waals physical adsorption to a magnetic catalyst surface [23]. Solid catalysts dramatically increase the conversion rate as the physical adsorption of H_2 facilitates the inhomogeneous magnetic torque to be applied to each atom of the H_2 molecule to induce spin state change [20]. The early Wigner theory [24] postulated that catalyst activity was proportional to the square of the catalyst magnetic moment. However this did not hold true for rare earth oxides and across different classes of materials [20, 21]. As such, the Wigner theory is recommended to only be taken as an indicator of catalyst activity but it does highlight the relation of catalyst activity to the number of magnetic sites. Since the Wigner theory development, non-dissociative O↔P catalysis has been demonstrated on various catalyst magnetic phases including diamagnetic [25], paramagnetic [25], ferromagnetic [26], and antiferromagnetic [27], with the magnetic phases illustrated in Figure 2. While the rates of conversion significantly differ from material to material, this observation indicates the P→O conversion is linked to the surface magnetic inhomogeneity rather than the bulk magnetisation of the material.

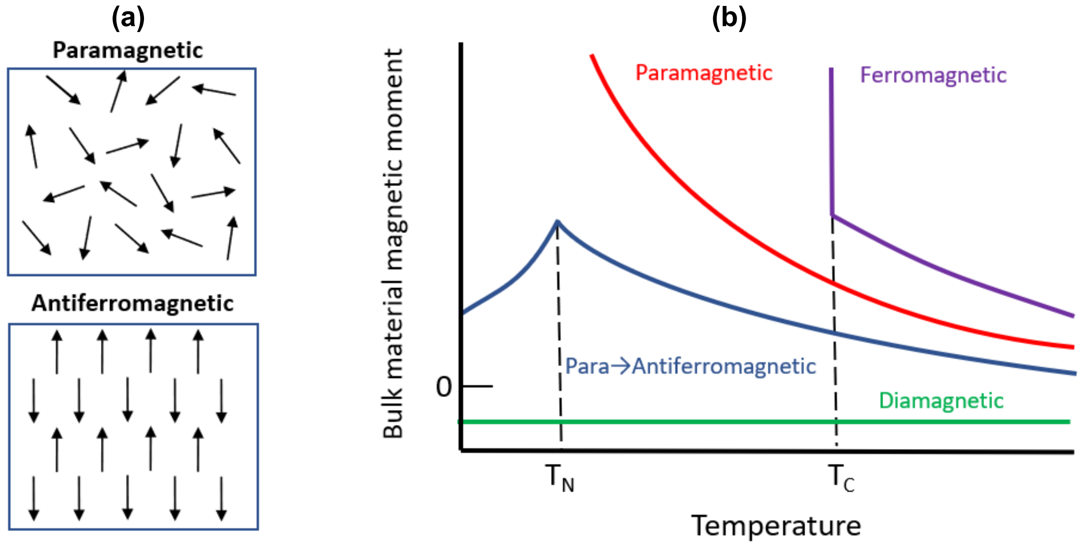


Figure 2: Illustration of magnetic structures. (a) Illustration of magnetic dipole orientation in paramagnetic and antiferromagnetic materials [28]. (b) Net magnetic moment of various magnetic structures as a function of temperature [29]. T_N denotes the Néel temperature where a paramagnetic material becomes antiferromagnetic upon cooling. T_C denotes the Curie temperature where a ferromagnetic material becomes paramagnetic upon heating.

It is possible that antiferromagnetic catalysts have a bulk antiferromagnetic phase with a paramagnetic outer layer due to unpaired magnetic dipoles on a rough catalyst surface. Varying degrees of paramagnetic and antiferromagnetic behaviour were exhibited on Cr_2O_3 particles of varying sizes [30] indicating the importance of the catalyst synthesis method and morphology on the surface inhomogeneous magnetic field. The dependence of catalyst activity on the surface magnetic inhomogeneity and the number of magnetic sites poses challenges for identifying desirable catalyst properties as magnetic moment characterisation of catalysts is conventionally a macro scale technique that cannot resolve the surface properties. As such, future catalyst material development may benefit from catalyst morphology characterisation alongside modelling the surface magnetic properties to relate catalytic observations to theory.

The inhomogeneous magnetic moment experienced by the adsorbed H_2 molecule is dependent on its orientation and thus the rotational energy level of the adsorbing H_2 species [23]. Studies experimentally observed o-H_2 selectively adsorbs to a catalyst surface over p-H_2 due to an anisotropy related to H_2 adsorption [31, 32]. This is attributed to the phenomena of hindered rotation where the physically adsorbed H_2 exhibits partial rotational energies between the discrete J = 0, J = 1… etc levels [23]. As adsorption is a key process of catalysis, the adsorption energy barrier and population of adsorption sites with inhomogeneous magnetic moments are indicated to be key factors in catalyst effectiveness [33].

Influencing factors on magnetic acceleration

As discussed above the activity of P→O catalysts appears linked to the population of active sites and an energy barrier of adsorption/surface spin conversion. This was based on the requirements of an inhomogeneous magnetic torque and the hindered rotation during H_2 physical adsorption to the catalyst surface. Catalyst activity is also linked to the surface magnetic properties which can be influenced by the magnetic phase and morphology of the material as well as through an externally applied magnetic field.

The extent of magnetic acceleration of the P→O conversion rate on the same catalyst has been observed to be influenced by varying pre-treatment conditions [19]. This is attributed to the accessibility of p-H_2 molecules to the paramagnetic catalyst sites and indicates that the magneto-catalytic effect to be dependent on overcoming an energy barrier as proposed in the theory discussed above. Magnetic acceleration was generally observed to be proportional to the strength of the magnetic field applied until saturation was reached where increasing the magnetic field strength further did not result in an increasing conversion rate [17]. The saturation magnetic field and the extent of magnetic acceleration at saturation varied significantly across catalyst materials but could be grouped into several archetypes [18]. This saturation effect suggested a link to the population of active sites where there would be a finite limit of magnetic acceleration once all active sites were influenced by the applied magnetic field. Ng and Selwood observed increasing magnetic deceleration until saturation was observed by increasing the applied field strength to Cr_2O_3 and CoO catalysts at their respective Néel temperatures [27]. Whereas dilute Cr_2O_3 on Al_2O_3 catalysts exhibited strong magnetic acceleration until saturation at temperatures far above and below their Néel temperature [34]. This suggests that analysing the magneto-catalytic effect based on macroscale magnetic properties may not be adequate, considering the conversion mechanism depends on local magnetic properties at the length scale of a hydrogen molecule.

The influence of surface magnetic properties on the P→O conversion was assessed through various theoretical treatments [20, 35-39]. Modelling indicated that alignment of the catalyst magnetic field parallel to the surface would increase the P→O conversion rate [37]. Further theoretical modelling by Ilisca [38] suggested a resonance mechanism where the applied magnetic field altered the electron band gap energy on the catalyst. This could either tune or detune the catalyst's electron band gap, bringing it closer to or further from the rotational energy gap of P→O conversion, thereby facilitating the energy exchange required for conversion. These theoretical relationships could support the above observations that the magneto-catalytic effect is both linked to the population of active sites and an energy barrier. In this case, the application of a magnetic field could influence the direction of the surface magnetic dipoles for closer adsorption and the population of active sites that have a tuned electron band gap to induce conversion.

The cumulative experimental and theoretical literature demonstrates that the magneto-catalytic effect is strongly influenced by the catalyst's chemistry, the magnetic field strength and operating temperature. Experimental observations suggest that the magneto-catalytic effect has a dual mechanism, involving both the population of active sites and the energy barrier of adsorption. However, limited kinetic analysis of experiments has been conducted to evaluate this. It is unclear if these observations were specific to the dissociative P→O conversion mechanism as the majority of experiments were conducted near ambient temperature while the non-dissociative conversion mechanism is dominant below T = 100 K [19, 23, 40]. The limited breadth of experiments has meant that no model has been developed to quantify the magneto-catalytic effect over conventional catalysts including ferric oxides at cryogenic temperatures. High-quality experiments would benefit the scientific community by enabling more rigorous evaluation of existing theories. Specifically, kinetic and uncertainty analyses could help quantify the effects of applied magnetic fields on the activation energy and active site population, while building confidence for its practical application. In this work, kinetic and uncertainty analysis have been applied to literature magnetic P→O catalysis data on Cr_2O_3 [34] to support comparison to new experimental magneto-catalytic observations.

Magneto-catalytic effect uncertainty

Selwood et al. [34] investigated magnetic acceleration of the P→O conversion over various Cr_2O_3 catalyst concentrations at temperatures from T = 123 – 373 K. The feed H_2 was converted to 0.505 p-H_2 fraction in a liquid nitrogen bath before entering a packed bed catalytic converter surrounded by an electromagnetic coil [41]. The P→O conversion was conducted with and without a magnetic field present and the converter outlet p-H_2 composition was measured to estimate the net molar conversion rate for the converter. The ratio of the molar conversion rate with and without the applied magnetic field was used as the definition of magnetic acceleration as in equation (1):

$$\text{Fractional magnetic acceleration} = \left(\frac{F(X_1 - X_2)_{\text{magnet}}}{F(X_1 - X_2)_{\text{baseline}}} - 1\right), \tag{1}$$

where F is the molar flow rate, X is the p-H_2 fraction, and 1 and 2 reference the inlet and outlet of the converter, respectively. While this expression may be appropriate as a general estimate of magnetic acceleration, it does not consider that the conversion rate is a function of the driving force from equilibrium (X - X_e, where X_e is the equilibrium p-H_2 fraction at that temperature) in the first order and Langmuir kinetic models. The first order kinetic model was applied to analyse the magnetic acceleration using the first order rate constant, k, to reflect catalyst activity rather than the molar conversion rate. The first order model was applicable for use with the literature data as only a single flow rate was utilised. Applying and re-arranging the first order kinetic model of equation (2) into equation (3) shows that changes in the outlet p-H_2 fraction are related to exponential changes in the catalyst activity, k, and this means that small differences in the magnet and baseline molar conversion rates could imply significantly different catalyst activities. This also means that the 0.01 p-H_2 fraction measurement uncertainty [41] propagates into much larger uncertainties in the rate constant though the uncertainty in the magnetic acceleration was not quantified in the original literature.

$$\frac{\mathrm{d}C}{\mathrm{d}t} = kC(X - X_\mathrm{e}) \tag{2}$$

$$k = -\frac{ln\left[\frac{(X_2 - X_\mathrm{e})}{(X_1 - X_\mathrm{e})}\right]}{t} \tag{3}$$

A more appropriate definition of the magneto-catalytic effect is the ratio of the catalyst activity (expressed by the first order rate constant, k) in equation (4) which is independent of the equilibrium driving force. The outlet p-H_2 fraction was calculated based on the molar conversion rates reported and was used in the integrated and rearranged form of the first order model of equation (3) together with equation (5) to find the baseline and magnet rate constants. The 0.01 p-H_2 composition measurement uncertainty was propagated through the baseline and magnetic rate constants to estimate the potential range of the magnetic acceleration via ASTM E2586 [42]. This was performed using (6) to represent the propagation of the p-H_2 measurement uncertainty to the rate constant based on equation (3). The literature and re-interpreted results are presented in Figure 3.

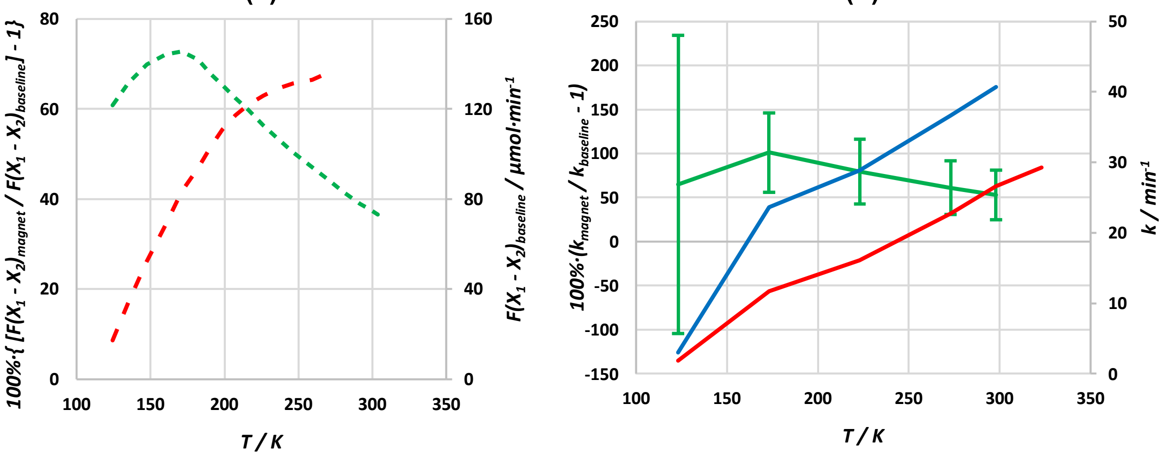
$$\text{Fractional magnetic acceleration } = \left(\frac{k_{\text{magnet}}}{k_{\text{baseline}}} - 1\right) \tag{4}$$

$$t = \frac{V}{\dot{v}} \tag{5}$$

$$\Delta\text{k} = \sqrt{\left[\frac{-\Delta X}{t(X - X_e)}\right]^2} \tag{6}$$

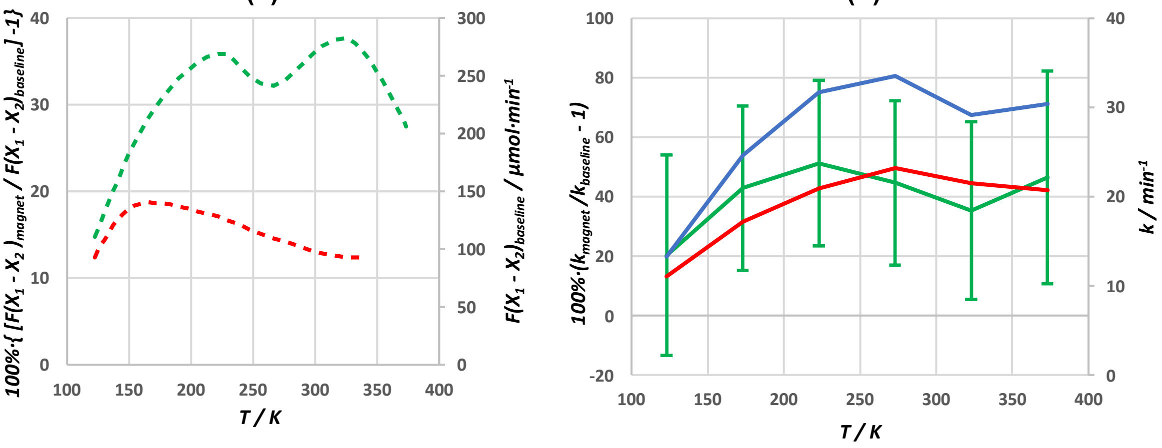
0.0028 % Cr_2O_3 on Al_2O_3 support at 7.5 kOe magnetic field strength

(a) **(b)**



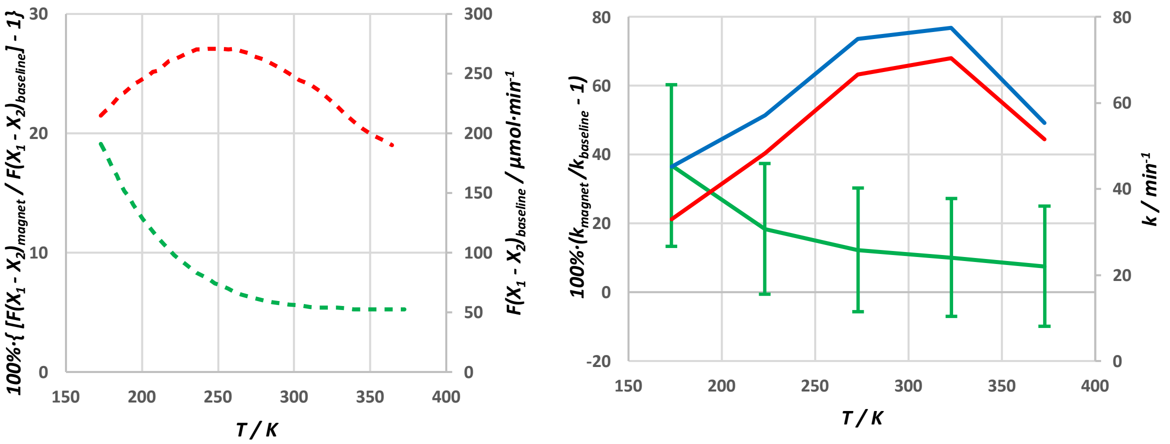
0.17 % Cr_2O_3 on Al_2O_3 support at 7.5 kOe magnetic field strength

(c) **(d)**



1 % Cr_2O_3 in Al_2O_3 solid solution at 7.5 kOe magnetic field strength

(e) **(f)**



α-Cr_2O_3 crystal at 7.5 kOe magnetic field strength

(g) **(h)**

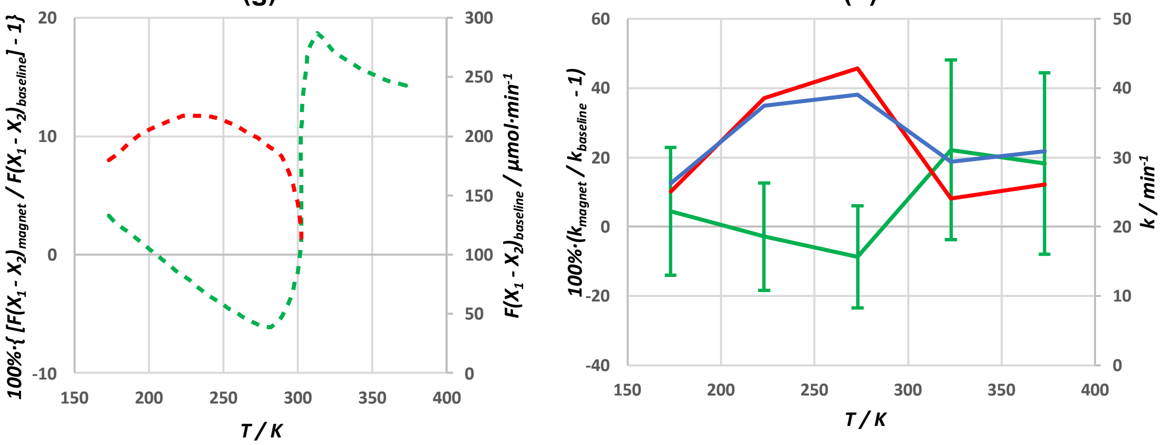


Figure 3: Uncertainty analysis of magnetic-acceleration on various Cr_2O_3 catalysts. (a), (c), (e), (g) were reproduced from literature [34]. (b), (d), (f), (h) are re-interpretations of the original data using the rate constant definition of magnetic acceleration. Curves: - - -, magnetic acceleration; - - -, baseline conversion rate; —, fractional magnetic acceleration; —, baseline rate constant; —, magnetic rate constant.

Defining magnetic acceleration based on rate constants generally resulted in higher magnetic acceleration values compared to the molar conversion rate basis. Additionally, comparing Figure 3 (c) and (d) for the magneto-catalytic effect on 0.17% Cr_2O_3, some behaviour such as the local minimum at $T \approx 263$ K is not observed. It is also uncertain if the small local magneto-catalytic acceleration maximum at $T \approx 173$ K on the 0.0028 % Cr_2O_3 sample is actually present based on the propagated uncertainty from the composition measurement. The 0.01 p-H_2 composition uncertainty resulted in an average rate constant uncertainty of 14% and an average fractional magnetic acceleration uncertainty of 32%. While the uncertainty range for all materials was large, the magneto-catalytic acceleration was statistically verified above $T = 150$ K for 0.0028% and 0.17% Cr_2O_3, though it could not be statistically verified for most temperatures for 1% and α-Cr_2O_3. The overall increasing and decreasing trends of the magneto-catalytic effect as a function of temperature were confirmed. A major contributor to the large uncertainty range was that the p-H_2 composition measurement uncertainty was 0.01 while the change in the p-H_2 fraction across the converter was approximately 0.05 to 0.15. Ideally, a larger p-H_2 fraction change across the packed bed would reduce the uncertainty but the constraints of placing the converter within an electromagnet likely resulted in the compromise of using a small converter.

The baseline activities of the Cr_2O_3 catalysts are compared in Figure 4. A significant change in catalyst activity is observed on α-Cr_2O_3 at the Néel temperature of $T = 307$ K [27]. This effect was also observed for a CoO catalyst but the activity was symmetrical around its Néel temperature [27]. The 1% Cr_2O_3 also exhibited a change in activity at the Néel temperature; however the material was expected to be weakly ferromagnetic based on magnetic susceptibility characterisation of the sample [18] and similar Cr_2O_3/Al_2O_3 solid solutions prepared in literature [43]. The peak activity around the Néel temperature could coincide with the resonance theory [38] where the magnetic reordering favourably aligns the catalyst electron band gap towards the P→O conversion energy. However, the activity change for the antiferromagnetic α-Cr_2O_3 and ferromagnetic 1% Cr_2O_3 at the Néel temperature may mean that the resonance effect is not solely based on a magnetic phase transition. Conversely, the dilute samples (<1% Cr_2O_3) did not appear to have a significant baseline activity peak for the temperature range tested. It was possible that their Néel temperature was outside of the temperature range tested or that they were paramagnetic due to a rough surface with unpaired dipoles as exhibited on other materials [30]. The magnetic properties of these catalysts were generally uncharacterised posing challenges for linking the activity to the magnetic properties.

Assuming that the 1% Cr_2O_3 sample was ferromagnetic and that the <1 % concentrations were paramagnetic would imply increased magnetic alignment at lower temperatures while the antiferromagnetic α-Cr_2O_3 would become overall less magnetic on a macro scale. The common decrease in activity at lower temperatures across all magnetic structures indicates the net magnetisation of the catalyst may not sufficiently describe the catalyst activity or the number of active sites. Rather, the catalytic behaviour is likely influenced by local magnetic inhomogeneity at the surface and the number of active sites, which may correspond to the number of inhomogeneous magnetic sites. It is possible that the common decrease in activity at lower temperatures could indicate other factors are involved in the conversion such as the rate of p-H_2 adsorption on the catalyst surface. In this case, the decreasing activity may reflect the increasing energy barrier of adsorption assuming an Arrhenius relationship. Baseline activity trends may also be influenced by catalyst morphology, either by directly affecting H_2 adsorption or by changing the surface magnetic properties. The temperature dependent activity of dilute Cr_2O_3 is similar to that observed in experiments on amorphous chromia gel catalysts, whereas the activity of α-Cr_2O_3 was consistent with results from tests on macro-crystalline Cr_2O_3 [44]. This further reinforces the idea that bulk magnetic property measurements may not be suitable for assessing the magneto-catalytic mechanism, given that activity is likely related to the inhomogeneity of the catalyst's surface magnetic fields at the length scale of H_2 molecules. Additionally, it remains unclear if the Néel temperature and crystal structure are of importance to conversion at cryogenic temperatures due to the shift in the conversion mechanism from chemisorption to physisorption [19, 40].

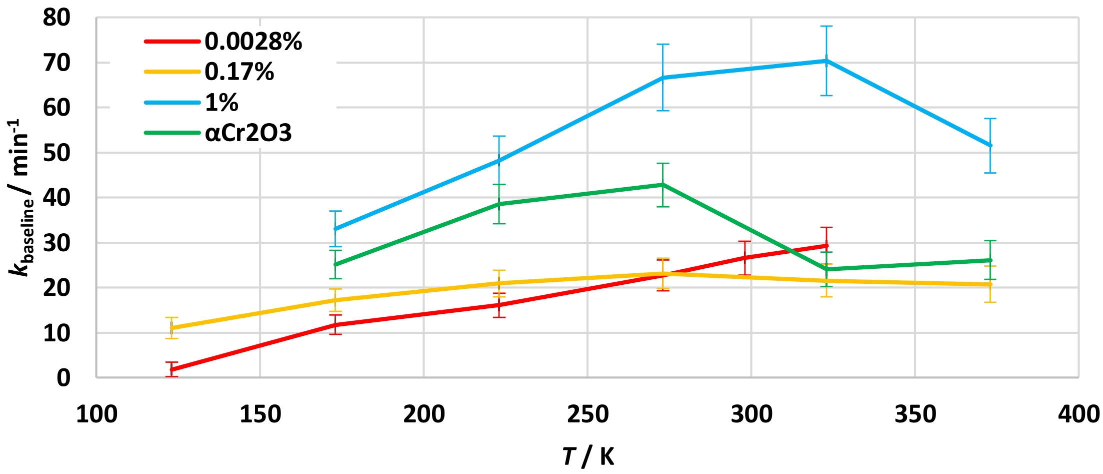


Figure 4: Baseline rate constants of various Cr_2O_3 catalysts with respect to temperature. Analysis is based on literature experiments [34].

The rate constant based magnetic acceleration behaviour of the Cr_2O_3 samples is presented in Figure 5 with uncertainty bars omitted for visual clarity. It is unclear if these catalysts were subject to the magnetic field as they were cooled down as this would impact the macro scale magnetic structure of the catalyst for the low temperature tests. If cooled within the magnetic field, the antiferromagnetic α-Cr_2O_3 may have maintained a net magnetisation below its Néel temperature, though it is unclear how it may have impacted the surface magnetic inhomogeneity. The dilute Cr_2O_3 samples appeared to have a magnetic acceleration peak at T = 223 and 173 K for the 0.17% and 0.0028% Cr_2O_3 samples, respectively, but the large uncertainty range meant it was difficult to verify this. It is also possible that magnetic acceleration increased as the temperature decreased similar to the ferromagnetic 1% Cr_2O_3 sample. The magnetic acceleration of the α-Cr_2O_3 sample appeared to follow similar behaviour to the baseline activity with a large fluctuation at the Néel temperature. Further assessment of the magnetic acceleration as a function of temperature is made difficult by limited magnetic property characterisation and the large uncertainty bounds based on the small extent of P→O conversion.

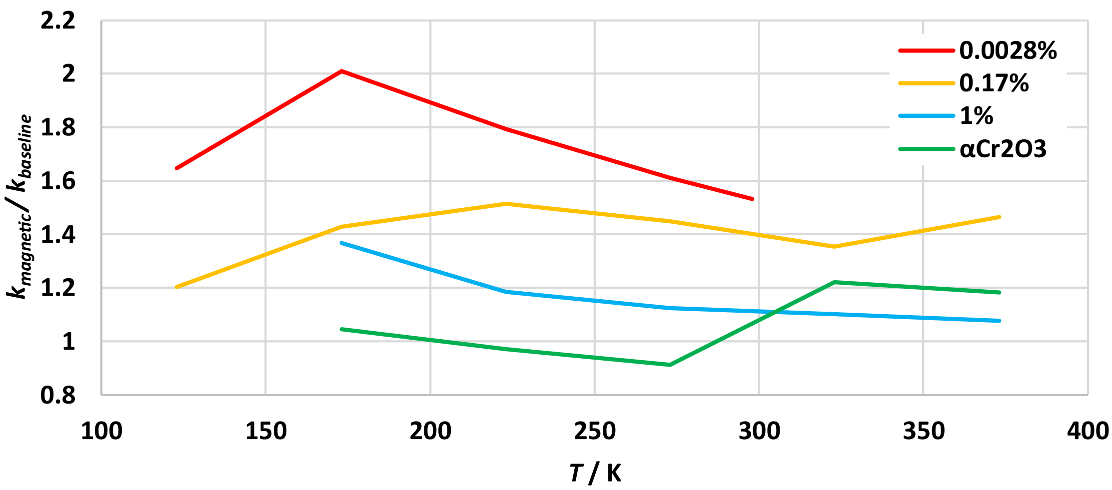


Figure 5: Magnetic acceleration on various Cr_2O_3 catalysts at a magnetic field strength of 7.5 kOe. Analysis based on literature experiments [34].

The complexity of the p-H_2 conversion phenomena highlights the need to complete further magneto-catalytic experiments in tandem with magnetic property characterisation and kinetic analysis to support the development of theory. Conducting magneto-catalytic studies over the conventional Ionex catalyst at cryogenic temperatures is of immediate interest to assess the utility of the effect for cooling SC motors. Kinetic data would be useful to evaluate changes to the number of active sites and activation energy with a magnetic field to support the development of theory on the P→O conversion mechanism. This paper focuses on experimental measurements of the magneto-catalytic effect on Ionex from a kinetics perspective to quantify the change in activity with an applied magnetic field. These results were linked to magnetic property characterisation to support future efforts in magneto-catalytic modelling and theory development. The results were then used to assess the applicability of magnetically accelerated P→O cooling for SC motors.

For the purposes of this study on progressing the technology readiness of P→O coolers, the commercial ferric oxide catalyst, Ionex, was selected for performing kinetic experiments with and without an applied magnetic field. P→O catalysis over Ionex was performed over temperatures of 45 to 90 K with an applied magnetic field strength of 2 kOe to reflect similar conditions to a P→O catalytic cooler installed adjacent to a superconducting motor. First order kinetic modelling was applied to quantify the magneto catalytic effect to estimate the practical implications of the effect on the required catalyst volume for P→O coolers. Kinetic implications of the magneto-catalytic experiments were linked to a bulk magnetic characterisation of Ionex powder to support future development of P→O conversion mechanism theory.

3. Methodology

The Cryocatalysis Hydrogen Experimental Facility (CHEF) [45] at the Hydrogen Properties for Energy Research (HYPER) Lab, Washington State University was utilised to assess the impact of an externally applied magnetic field on the activity of Ionex ferric oxide catalyst. This was conducted by measuring the change in p-H_2 fraction across an isothermal catalyst packed bed at various steady state conditions with and without an applied magnetic field. The experimental data was then used to quantify the catalyst activity represented by the first order rate constant. Derivative kinetic parameters, such as the pre-exponential factor and apparent activation energy, were then calculated to assess the impact of an applied magnetic field on the number of active sites and energy barrier of conversion.

Isothermal P→O converter and control

A simplified experimental schematic is highlighted in Figure 6. Further details of the apparatus are described by Bunge [45]. Each experiment involved the accumulation of a batch of LH_2 (0.998 p-H_2 fraction) utilising a cryocooler and an Ionex packed bed. During testing, the LH_2 was vaporised via a heater before flowing through the isothermal catalyst converter and a downstream mass flow controller.

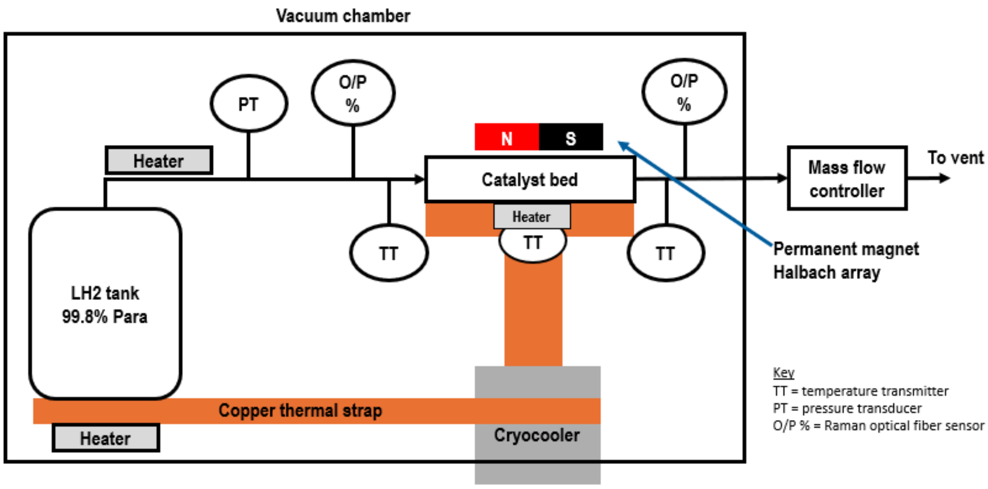


Figure 6: Experimental schematic diagram.

P→O conversion was conducted at the steady state points in Table 1 with and without the external magnetic field applied. The baseline (no magnet) and applied magnet tests were repeated twice at each steady state point for the same catalyst sample and activation to verify reproduceable results. The pressure was attempted to be kept constant, but did vary between P = 4.5 – 7.5 Bara across the range of steady state points due to the complexity of boiling LH_2 while simultaneously controlling the p-H_2 feed temperature and flow rate with heaters. H_2 flow rates were selected to achieve outlet p-H_2 compositions that were neither too close to equilibrium or the inlet composition to assist estimation of catalyst activity. The inlet and outlet p-H_2 composition was measured utilising Raman spectroscopy with the specific instruments and analysis method previously reported [46]. For all test cases, the inlet p-H_2 composition was measured to be within the uncertainty range of 0.998 p-H_2. Each steady state point was held for 1 minute for composition analysis via the Raman system. Details of the experimental control and measurement systems are expanded upon in Table 2.

Table 1: Steady state operating conditions.

T / K	$\dot{v}_{STP}$ / L·min^{-1}	P / Bara
45, 60, 90	1, 2, 4	4.5 – 7.5

Table 2: Details on experimental control and measurement.

Experimental features	Details
Mass flow rate control	Controlled by an Alicat mass flow meter (laminar differential pressure type) with hydrogen gas flow at ambient temperature to minimise uncertainty due to Para/Ortho property deviations.
Pressure control	Manually controlled by changing the heater power applied to the liquid hydrogen tank
Isothermal Para-Ortho converter bed temperature control	Controlled by cartridge heater in a control loop with Lakeshore Cernox® temperature sensor fixed at the midpoint of the converter length. An in-line Cernox® temperature sensor was installed upstream and downstream of the converter to verify isothermal operation. The feed hydrogen was heated to the converter temperature via the cartridge heater at the outlet of the liquid hydrogen tank.
Ortho-Para composition analysis	Measured via Raman spectroscopy with a 1-minute sampling time. Raman spectrum analysis via code with an estimated p-H_2 composition uncertainty of 0.015 [46].

Catalyst converter

The P→O converter was a stainless steel 316 tube with an inner diameter of 3.34 mm and length of 129.5 mm. The middle 40% of the tube volume was packed with 0.56 g of Ionex supplied by Molecular Products [13] while the top and bottom sections of the tube were each filled with 0.58 g of Al_2O_3 sourced from Atlantic Equipment Engineers [47] as a pseudo-inert filling that does not exhibit a magneto-catalytic effect [19]. The Al_2O_3 particles were anticipated to have minimal surface area for catalysis as they were made by fusion of alumina in an arc furnace as compared to the crystallisation process utilised in high surface area ferric oxide production. While the surface area and catalytic activity of the Al_2O_3 sample was not measured, literature experiments found that an α-Al_2O_3 sample with 9.2 m^2/g had a first order rate constant of 0.395/min [48], which is much lower than the 188/min first order rate constant obtained over a ferric oxide catalyst prepared via crystallisation [49]. Consequently, the catalysis over the Al_2O_3 was assumed to be negligible compared to the Ionex. Filling the center section of the tube with Ionex was done to ensure a close to homogenous magnetic field strength across the catalyst bed as discussed further in the subsequent section. Both the Ionex and Al_2O_3 had a particle size distribution fitting a 30 – 50 mesh (300 – 600 μm particle diameter) to minimise the possibility of mixing between the powder layers.

The converter tube was shrouded in a copper thermal strap in contact with a cryocooler and heater to maintain an isothermal converter temperature during the endothermic P→O conversion. The catalyst was activated at T = 373 K in a vacuum environment of approximately P = 10^{-2} torr for 4 hours prior to testing. This activation was selected to remove surface moisture from the catalyst without altering the catalyst chemistry. It was noted that other activation conditions have been tested in literature and could offer enhanced activity through changing the catalyst chemistry [50].

Magnetic Halbach array

The magnetic field was supplied through a permanent magnet Halbach array which offered a compact size and could be easily installed within the existing experimental apparatus. Five neodymium cube magnets of grade Nd52 and 25.4 mm side length [51] were arranged in the Halbach orientation shown in Figure 7 to produce a constructive magnetic field in the vicinity of the catalyst bed. The magnets were assembled into an aluminum tray and mounted parallel to the catalyst bed with a distance of 15 mm between the surface of the Halbach array and the converter centerline. The magnetic field was simulated in Ansys Electronics finite element analysis software to assess the magnetic field strength along the length of the converter.

As shown in Figure 7, the calculated magnetic field strength decreased by 50 % near the ends of the magnet array whereas the middle 40 % of the converter experienced a relatively consistent magnetic field strength. The middle 40 % of the converter length was packed with Ionex while the outer edges were packed with pseudo inert Al_2O_3 spacer to achieve a uniform magnetic field strength over the Ionex catalyst. The diamagnetic Al_2O_3 was not anticipated to appreciably impact the magnetic flux in the converter.

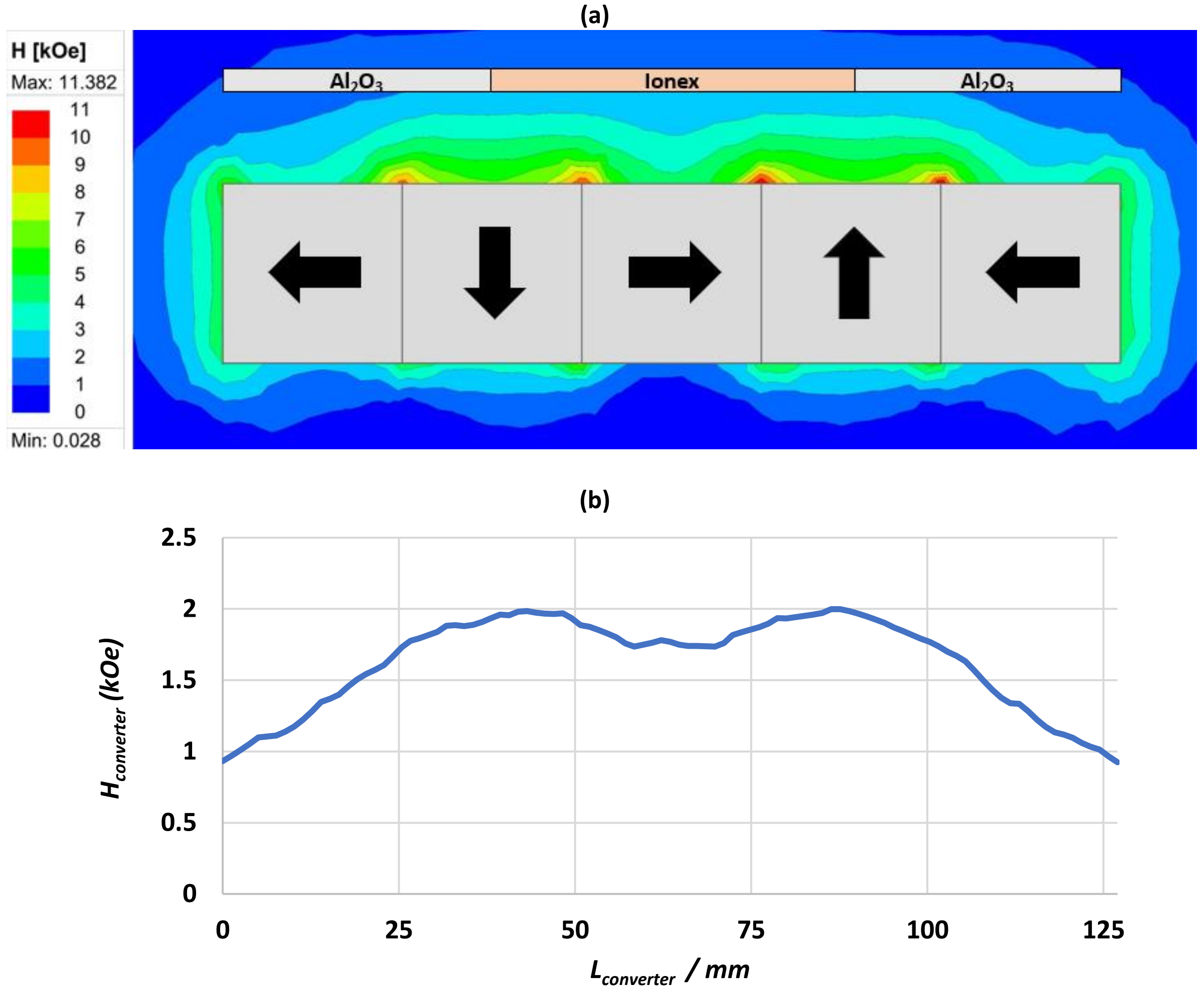


Figure 7: Magnetic field strength along P→O converter. (a) Magnetic field strength, H_{mag}, of the halbach array adjacent to the P→O converter. Modelled with Ansys finite element analysis. (b) magnetic field strength experienced at the centreline of the converter.

While the achieved field strength of 2 kOe was lower than the 7.5 kOe magnetic field strength applied in literature [34], it was deemed sufficient to assess if a magneto-catalytic effect was exhibited with Ionex as the literature displayed a consistent trend of increasing magneto-catalytic effect with increasing applied magnetic field strength until saturation. The applied strength was deemed to be practically relevant as megawatt scale superconducting motors typically have a stator magnetic field flux of 1 T corresponding to a surface magnetic field strength of 10 kOe [1, 7]. This would suggest that a P→O converter in the vicinity could experience a 2 kOe magnetic field strength in a practical application.

The permanent magnet tray was installed adjacent to the catalytic converter using an aluminum thermal strap mounted to the cryocooler as shown in Figure 8. The magnet array was installed prior to cool down of the system and meant that the Ionex catalyst was cooled in a magnetic field. It is unclear from literature if the Selwood experiments [17] featured field cooling of the Cr_2O_3 as the electromagnet utilised in their system could be turned on during, or alternatively only after, cooldown. The use of the thermal strap allowed the magnet array to reach thermal equilibrium with the system. While the temperature of the magnets within the tray could not be ascertained, literature characterisation of Nd52 grade magnets indicates their strength to remain within 5% of their room temperature behavior when cooled to the experimental operating temperatures of T = 45 – 90 K [52].

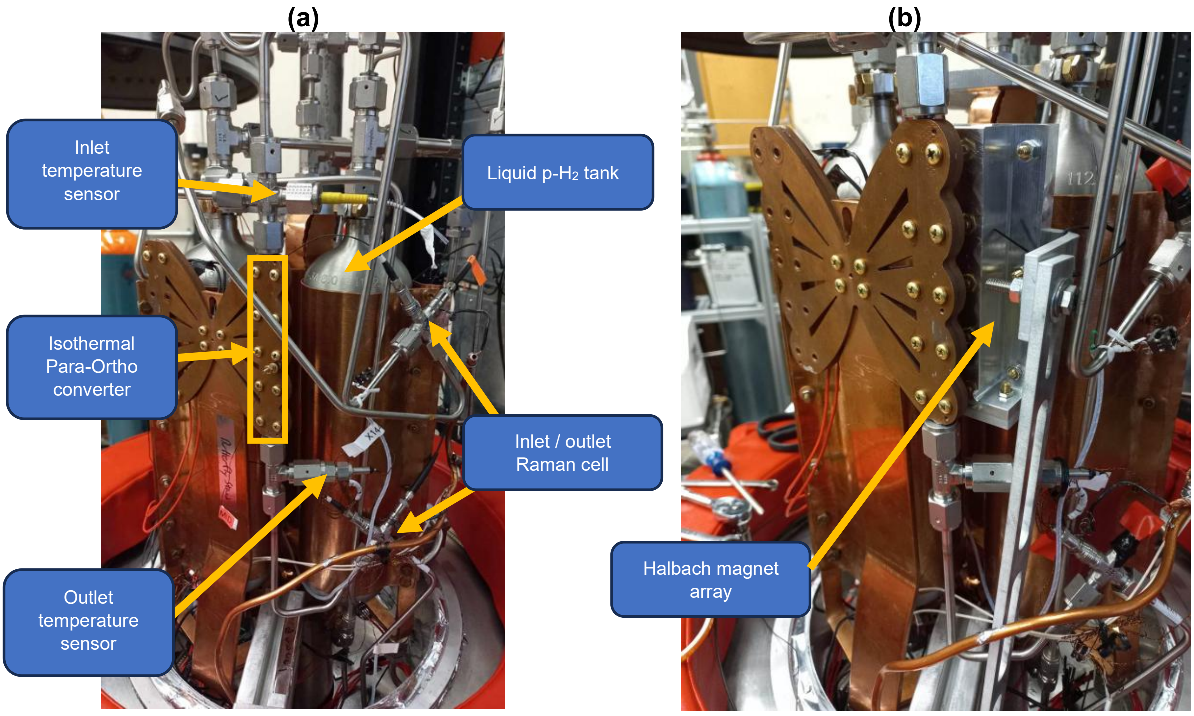


Figure 8: P→O converter experimental configuration. (a) baseline experimental configuration. (b) Halbach magnet installed parallel to the packed bed converter.

Kinetic analysis

The definition of magnetic acceleration is described by equation (7) as the ratio of the Ionex rate constant with the magnetic field applied compared to the baseline (no applied magnet) scenario.

$$magnetic-catalytic\ acceleration\ (MC) = \frac{k_{\mathrm{magnet}}}{k_{\mathrm{baseline}}} \quad \textbf{(7)}$$

This required an expression for the rate constant as a function of temperature and pressure to allow the fair comparison of catalyst activity at the same conditions. The general steps of the kinetic analysis were as follows:

1. Select kinetic model and calculate experimental rate constants for each steady state point.
2. Develop an expression for the experimental rate constant as a function of temperature and pressure.
3. Calculate the baseline and magnet rate constant for the same operating temperature and pressure to determine the extent of magnetic acceleration.
4. Perform uncertainty analysis via ASTM E2586 [42] by propagating the uncertainty in the p-H_2 fraction through the rate constant and then to the magnetic acceleration.

The Ionex rate constant was defined based on the first order kinetic model of equation (3) which lumps all the steps of the P→O conversion (diffusion, adsorption, surface conversion) into a single rate. Previous first order kinetic modelling of literature ferric oxide conversion [40, 49, 53] by the authors predicted the outlet p-H_2 fraction with an average absolute deviation of 0.01 from experimental values and was deemed suitable for application in this study. The first order model was rearranged to the form shown in equation (8) and was applied to Ionex to obtain the experimental rate constant for each steady state point using the definition of time in equation (5). Equilibrium p-H_2 fractions (X_e) were calculated as per Eisenhut and Haberstroh [16].

The simplicity of the first order model comes with the trade-off of having challenges predicting the rate at different space velocities [53]. Consequently the first order rate constant appears as a function of the residence time despite the obvious suspect of diffusion mass transfer limitations not being present [40]. As such, the average first order rate constant was calculated across the 1, 2, and 4 SLPM flow rates at each steady state temperature. The average rate constant was solved iteratively by minimising the deviation of the outlet p-H_2 fraction calculated from the first order model ($X_{2,\text{model}}$) compared to the experimental outlet p-H_2 fraction ($X_{2,\text{expt}}$). Equation (8) was used to calculate the outlet p-H_2 fraction for an assumed rate constant. The average rate constant was then found by minimising the sum of squared residuals (SSR) given by equation (9) using the GRG non linear algorithm.

$$X_{2,model} = X_\text{e} + (X_1 - X_\text{e})\text{e}^{-kt} \tag{8}$$

$$SSR = \sum \left(X_{2,expt} - X_{2,\text{model}}\right)^2 \tag{9}$$

Due to slight variations in the temperature and pressure of the baseline and magnetic rate constants, it was not possible to directly compare measured values. Instead, comparison was made by expressing the rate constant as a function of temperature and pressure using the Arrhenius rate constant of equation (10). The basis for using this expression stems from the authors' experience in modelling ferric oxide activity of literature experiments, where it yielded an average p-H_2 fraction deviation of 0.02.

$$k = \frac{A}{P} e^{\frac{-E}{RT}} \tag{10}$$

The pre-exponential factor, *A*, and apparent activation energy, *E*, parameters were obtained by minimising the residual sum of squares between the average experimental rate constants and the Arrhenius rate constant expression. This process was conducted for the repeat tests of the baseline and applied magnet measurements to allow comparison of their activity as a function of operating conditions. The uncertainty in the p-H_2 fraction of 0.015 was propagated through these calculations to obtain an uncertainty estimate for the activity and kinetic parameters for each test. The uncertainty in the kinetic parameters of the baseline and magnet scenarios was then propagated to the magneto-catalytic effect using equation (11) based on the ASTM E2586 method [42].

$$\frac{\Delta MC}{MC} = \frac{\Delta k_{mag}}{k_{mag}} + \frac{\Delta k_{base}}{k_{base}} \tag{11}$$

4. Results

The first order kinetic parameters for the baseline and magnet P→O conversion experiments are reported in Table 3 with the raw experimental data presented in the supporting material. Average values for the kinetic parameters were solved by taking the arithmetic mean of the rate constants for the baseline and magnetic tests, respectively. The baseline Ionex tests had distinctly higher pre-exponential factor and activation energy values compared to the applied magnet tests and indicated that the applied magnet caused a statistically significant change in the catalyst kinetic properties. The changes in both kinetic parameters on the application of the external magnet could indicate a dual mechanism to the magneto-catalytic effect.

Decreasing the pre-exponential factor could suggest a decrease in the population of active sites while the decrease in activation energy could indicate a decrease in the energy barrier of hydrogen adsorption or spin conversion on the catalyst surface. Further commentary on the kinetic changes is presented in the discussion based on Ionex bulk magnetic characterisation. A major limitation on further mathematical kinetic analysis of the magneto-catalytic effect is the lumped nature of the first order model as the calculated activation energy may only represent an apparent activation energy of P→O conversion as compared to individual activation energies of adsorption and nuclear spin change. Future magneto-catalytic kinetic research would benefit from employing higher complexity kinetic models such as the Langmuir model, but this would require larger testing sample sizes and higher accuracy p-H_2 composition measurements.

Table 3: Kinetic parameters of baseline and applied magnet scenarios

	A / bara·min^{-1}	*A* Range / bara·min^{-1}	*E* /J·mol^{-1}·K^{-1}	*E* Range /J·mol^{-1}·K^{-1}
Ionex 1	3190	3588 - 2782	678	821 – 522
Ionex 2	3790	4404 - 3223	939	1105 – 769
Average Ionex	3387	-	787	-
Magnet 1	2267	2498 – 2040	512	654 – 362
Magnet 2	2072	2398 - 1751	507	675 - 327
Average magnet	2169	-	510	-

The relative activity of each experimental run was compared by substituting the kinetic parameters of Table 3 into equation (10) and shown in Figure 9. This comparison was made at an absolute pressure of 2 Bara however, the relative activity comparison would be consistent at other pressures as the rate constant was inversely proportional to pressure. The repeat tests of the applied magnet scenario achieved a more consistent grouping than the baseline Ionex tests. Deviation of the repeat runs could be impacted by the pressure fluctuations within each steady state point, the approximations of the first order kinetic model and potential impurities in the feed H_2 stream, though there was insufficient data to quantify these factors.

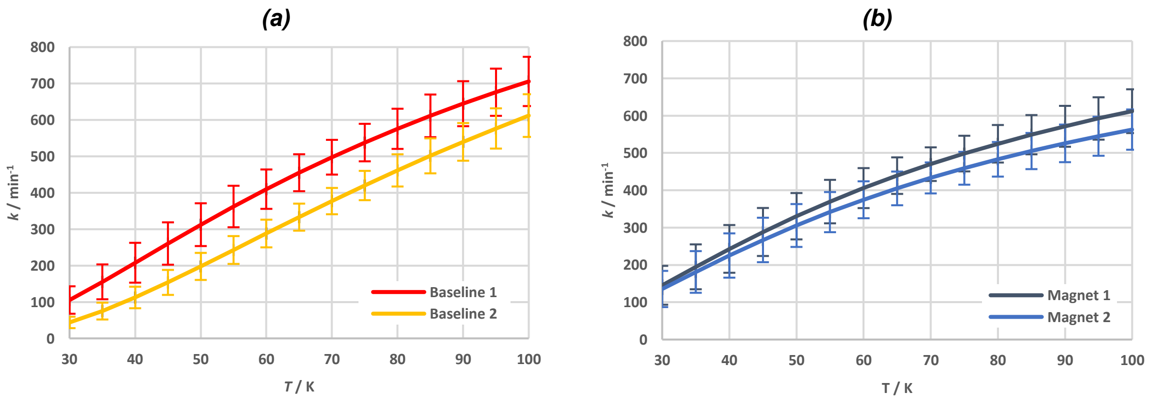


Figure 9: Plot of measured catalyst activity with temperature at 2 Bara. (a) Baseline activity. (b) Activity with an applied magnet.

Validity of the first order model

The deviations between the experimental and the first order model p-H_2 fraction at the outlet of the converter are plotted as a function of experimental p-H_2 fraction in Figure 10. On average, the first order model p-H_2 fraction deviated by 0.0157 from the experimental values. This was considered a suitable agreement for the purposes of this study given that the experimental uncertainty in the p-H_2 fraction was 0.015. The model overpredicted the activity at low flow rates and underpredicted the activity at high flow rates leading to several outliers with model deviations in excess of 0.02. this is primarily due to the simple assumptions of the first order model as also observed in literature [53].

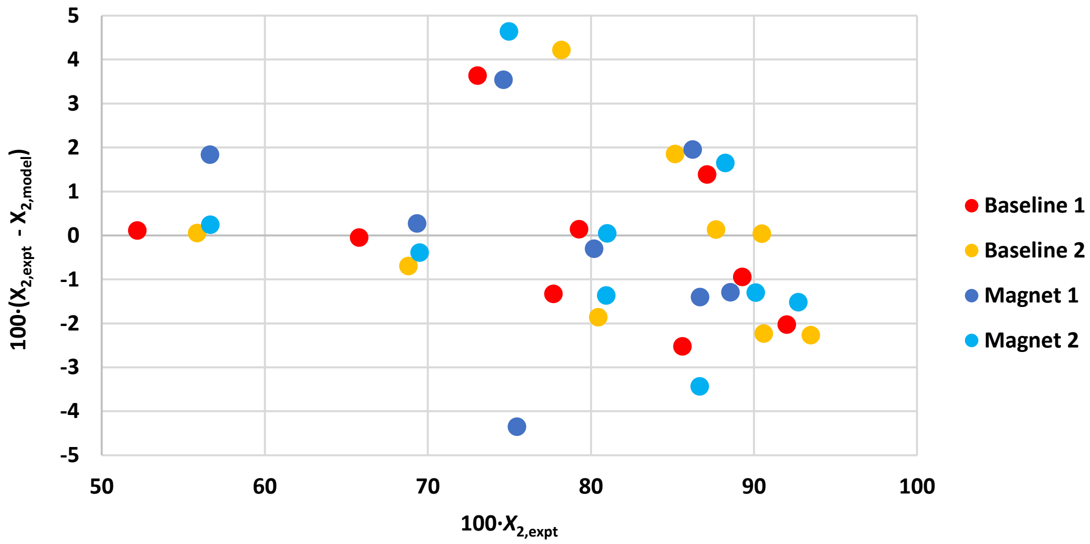


Figure 10: Parity plot of the first order model predicted p-H_2 fraction to experimental results.

Assessment of magneto-catalytic effect

The ratio of the magnetic and baseline average rate constants was then taken to assess the effect of the magnet on the catalyst activity in Figure 11. The results indicate that the applied magnetic field reduced the catalyst activity at T = 90 K but caused increased activity at T < 75 K, though the presence of a magneto-catalytic effect could not be statistically confirmed. From a kinetic perspective, this shift between the indicated magnetic acceleration and deceleration was caused by the reduction in the apparent activation energy of the conversion. The first order rate constant is proportional to the required catalyst volume and indicates that an applied magnet could reduce the required catalyst volume by nearly 50 % for cooling applications at T = 40 K. The increasing magnetic acceleration uncertainty at lower temperatures was primarily caused by the small p-H_2 fraction changes across the converter relative to the measurement uncertainty. Increasing the certainty of magnetic acceleration measurements at cryogenic temperatures would require higher accuracy p-H_2 fraction measurements such as those based on speed of sound-based systems [16]. The industrial applicability of magnetically accelerated P→O cooling is dependant on evaluating the reduction in catalyst volume required for P→O conversion/cooling with respect to relevant operating conditions.

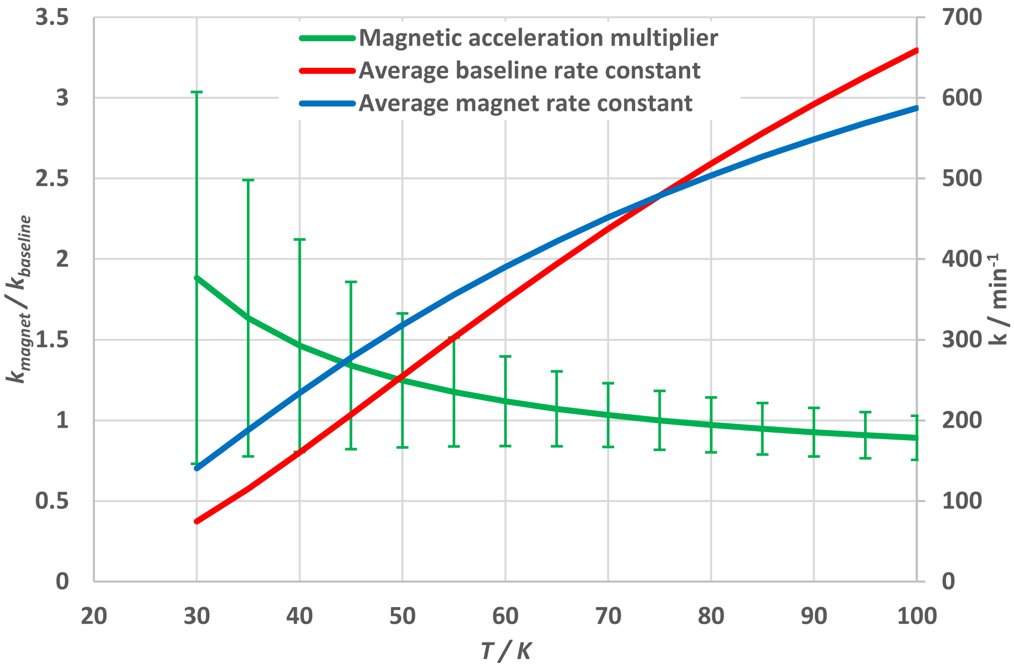


Figure 11: Magneto-catalytic effect on Ionex.

Implications for P-O cooling

The impact of the magnetic enhancement on the required catalyst volume for P→O coolers was calculated for various operating conditions using the first order kinetics model. Equations (3), (5) and (12) were combined and rearranged to give the required catalyst volume per hydrogen molar flow rate in equation (13). Equation (14) was used to describe the extent of P→O conversion and was substituted in to equation (13) to allow comparison of the required catalyst volume at different temperatures on a fair basis. A typical extent of 95 % conversion was selected as a generally applicable balance between catalyst volume and the change in p-H_2 fraction across a catalytic bed. The concentration of hydrogen was approximated by the ideal gas equation based on the expected LH_2 aircraft fuel system operating pressure of 2 Bara [54, 55]. The substitution of the rate constant of equation (10) made the required catalyst volume independent of pressure due to the inverse pressure dependency of the rate constant. The resultant equation (15) was then used to calculated the required catalyst volume for the baseline and applied magnet scenarios based on the kinetic parameters of Table 3.

$$\dot{v} = \frac{F}{C} \tag{12}$$

$$\frac{-ln\left[\frac{(X_2 - X_e)}{(X_1 - X_e)}\right]}{Ck} = \frac{\mathrm{V}}{F} \tag{13}$$

$$\varepsilon = \frac{|X_1 - X_2|}{|X_1 - X_e|} = 95\% \tag{14}$$

$$\frac{-ln(1-\varepsilon)}{RTAe^{\frac{-E}{RT}}} = \frac{V}{F} \tag{15}$$

Figure 12 shows that the required volume of catalyst in the baseline scenario increases very substantially below a temperature of 60 K. This was primarily caused by a significant decrease in the catalyst activity with temperature. The required catalyst volume gradually increases above T = 100 K due to the rate constant approaching an asymptote while the concentration of the hydrogen decreased linearly. Magnetic acceleration decreased the required volume proportionally to the magneto-catalytic ratio of Figure 11. This resulted in an approximately 50 % reduction in the required catalyst volume for cooling at T = 40 K. The lower activation energy associated with the presence of the magnetic field meant that the required catalyst volume was approximately independent of temperature for a given extent of conversion for cooling applications above 40 K. However, the outlet p-H_2 fraction ($\underline{X}_2$) and hence magnitude of energy removed would still be a function of temperature due to how the equilibrium p-H_2 fraction (Xe) changes with temperature.

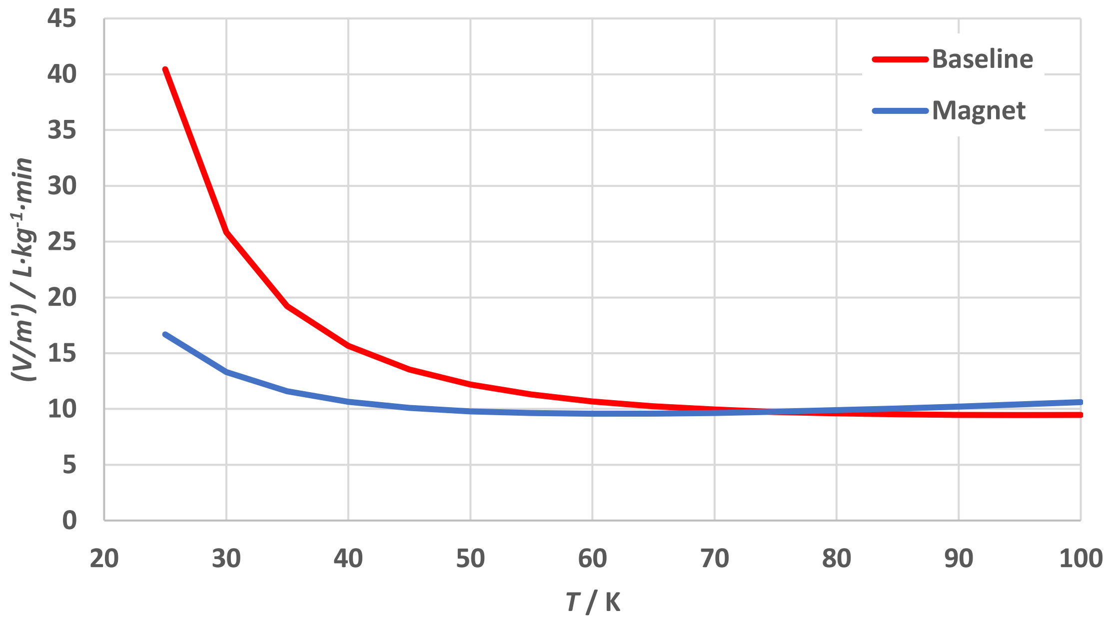


Figure 12: Comparison of required Ionex volume for parahydrogen conversion with and without an applied magnetic field.

The high-level trade-off between additional cooling duty and catalyst volume requirements for magneto-catalysis cooling was quantified based on a 2 MW superconducting motor, similar in scale to the engines used on a De Havilland Dash 8-300 [56]. A 2 MW motor would require approximately 2 kg/min of hydrogen flow to a fuel cell operating at 50 % efficiency and 4 kW cooling at T = 40 K [12]. The cooling duty of using the sensible heat of evaporated liquid hydrogen was compared to the cooling duty and volume requirements of a catalytic heat exchanger using both the sensible heat and P→O conversion. The sensible cooling component was calculated using the enthalpy balance of equation (16) with an inlet temperature of 22 K and outlet temperature of 40 K. The inlet temperature was based on the saturation temperature of p-H_2 at an absolute pressure of 150 kPa (T_{sat} = 21.6 K) which was found to be suitable for liquid hydrogen fuelled aircraft [54]. An extent of conversion of 95 % was assumed for the P→O cooling contribution giving an outlet p-H_2 fraction of 0.893 which was used in the enthalpy balance of equation (17).

The sensible heat component could provide 6.69 kW of cooling at T = 40 K whereas inducing the P→O conversion would increase the delivered cooling by 36 % to 9.14 kW. While the evaporated liquid hydrogen has sufficient sensible energy to meet the cooling needs, inducing the P→O conversion could offer value by providing cooling redundancy, critical for maintaining a superconducting state. The additional P→O cooling could also be valuable when the plane is idle at an airport where the nominal LH_2 fuel flow rate is not needed by the engines though further understanding of the passive heat leak to the motors would be required to evaluate the fuel saving. The P→O cooling contribution in this example represents a conservative case as storing LH_2 at elevated pressures would result in a higher inlet temperature and greater P→O cooling contribution relative to the decreased sensible cooling duty.

$$Q_{\text{sensible}} = \dot{m}\left[H_{\text{P}(T=40\ K)} - H_{\text{P}(T=22\ K)}\right] \tag{16}$$

$$Q_{\text{sensible+PO}} = \dot{m}\left[X_P H_{P(T=40\ K)} + (1 - X_P)H_{\text{O}(T=40\ K)} - H_{P(T=22\ K)}\right] \tag{17}$$

Inducing P→O cooling in a 2 kg/min LH_2 flow would require approximately 30 L of Ionex catalyst (bulk volume) based on Figure 12 and assuming isothermal operation at T = 40 K. Whereas installing the catalytic heat exchanger in the vicinity of the superconducting motor using the magneto-catalytic effect could reduce the catalyst volume by 10 L or the mass by 12 kg based on the catalyst packed bed density of 1.2 g/cm^3 [13]. Additional mass savings may result from using a smaller heater exchanger and cold box size, which are likely to be manufactured from materials such as Al6062 with a density of 2.7 g/cm^3 [57]. While the magnitude of volume saving via magnetic acceleration is small, the effect could be the difference between P→O cooling fitting or exceeding the mass and volume budget of an aircraft cryogenic heat management system. The estimated catalyst volume likely under-represents the required catalyst volume needed in a real system where an axial temperature gradient along the catalytic heat exchanger is present. A more detailed assessment of the P→O cooling duty and volume requirement should be undertaken using a combined kinetic and heat transfer model to discretise the axial temperature of the catalytic heat exchanger as the required catalyst volume increases significantly for T < 40 K.

Magnetic structure of Ionex

The magnetic susceptibility of Ionex was characterised as a function of temperature to attempt to relate its macro-scale magnetic structure to the magneto-catalytic results. A sample of Ionex was pre-treated under vacuum for 4 h at T = 140 °C and the magnetic susceptibility measured with and without the presence of an applied magnetic field was shown in Figure 13. The results suggest the material is antiferromagnetic with a Néel temperature of approximately 100 K. Application of the applied magnetic field during cooling resulted in a preservation of the net magnetic moment below the Néel temperature and suggests that the Ionex in the packed bed converter maintained a net magnetisation during the experiment. The increasing baseline activity of Ionex as a function of temperature could corroborate literature observations that a maximum in the kinetic activity is observed at the Néel temperature [27]. However, there is insufficient Ionex kinetic data above T = 100 K to evaluate this hypothesis. It is also uncertain if the Arrhenius kinetic model is applicable to describe the catalyst activity above T = 100 K if a potential maximum activity at the Néel temperature is present.

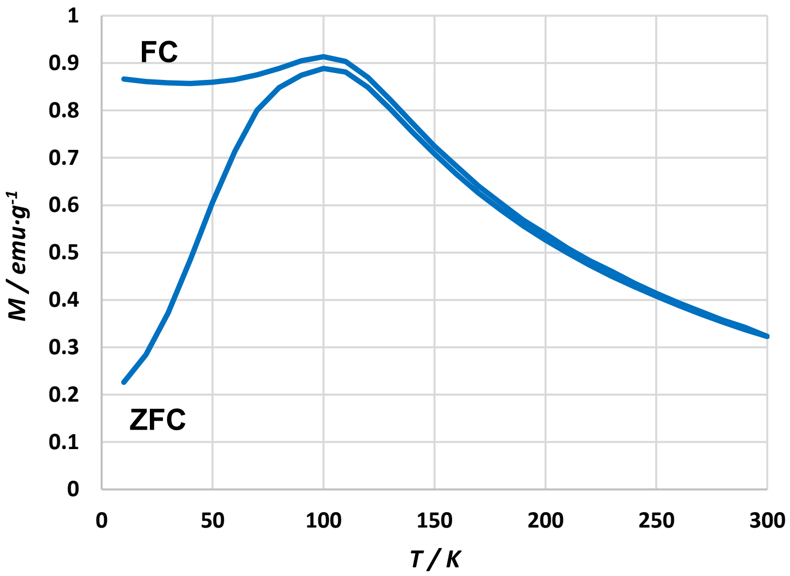


Figure 13: Specific magnetisation of Ionex pre-treated under vacuum for 4 hours at 140 °C. FC (field cooling) represents magnetisation during cooling with a 1 kOe applied field strength. ZFC (zero field cooling) represents magnetisation under no applied magnetic field. The Boskovic research group at the University of Melbourne is acknowledged for conducting this characterisation.

The requirement to cool superconducting motors prior to turning them on would indicate that the magnetic properties of the Ionex catalyst would be represented by the zero field cooled magnetic susceptibility test in practice. This would indicate that the catalyst would be more antiferromagnetic than the field cooled catalyst in this experiment but it is unclear how field cooling and zero field cooling impact the magnetic acceleration and further experiments to investigate this are recommended.

5. Discussion

The activity of the baseline and magnetic scenarios were appropriately described by the Arrhenius equation based on the experimental uncertainty. However, uncertainty propagation of the p-H_2 measurement led to large uncertainties in the extent of magnetic acceleration and posed challenges in confirming its practical value. The increasing uncertainty of catalyst activity at lower temperatures was primarily caused by the small change in p-H_2 fraction across the converter relative to the p-H_2 fraction measurement uncertainty. Reducing the uncertainty would require higher accuracy measurements of the p-H_2 fraction and a larger number of repeat experiments to verify average baseline and magnet kinetics. Conducting magneto-catalytic experiments on the O→P conversion at cryogenic temperatures could present an alternative path to reduced uncertainty assuming magnetic acceleration applies to both conversion directions. The O→P conversion direction has a larger change in composition across the converter at cryogenic temperatures and would reduce the impact of the composition measurement uncertainty.

While not necessarily mechanistic, the first order kinetic model suggests that the applied magnetic field reduced the number of active sites and the energy barrier of adsorption / surface spin conversion. This coincides with an increased net magnetisation of the Ionex during field cooling and suggests that maintaining a net magnetisation of the catalyst results in reduced magnetic inhomogeneity but increased hydrogen adsorption ability. It is unclear how increasing the applied field strength impacts the magnetic acceleration experienced on Ionex if at all, as its antiferromagnetic structure has low magnetic susceptibility. The saturation effect on paramagnetic and ferromagnetic Cr_2O_3 indicated that increasing the net magnetisation of the catalyst increases the magnetic acceleration [17]. Assuming the first order model is mechanistic, this implies a decreasing number of active sites but easier adsorption with increasing magnetic field strength.

The increasing magnetic acceleration of Ionex with decreasing temperature is similar to the magnetic acceleration observed in the literature for both antiferromagnetic α-Cr_2O_3 (below its Néel temperature) and ferromagnetic 1 % Cr_2O_3. Similar magneto-catalytic behaviour for different magnetic structures indicates that linking the magneto-catalytic mechanism solely to the bulk magnetic structure of the catalyst is not sufficient to explain the phenomena. The switch from magnetic deceleration to acceleration at T = 75 K could indicate similar behaviour to that observed on α-Cr_2O_3 where magnetic deceleration was largest immediately below the Néel temperature but becomes magnetic acceleration at lower temperature. A possible explanation of this could be the resonance mechanism described by Ilisca [38] but further understanding of the impact of field cooling on the surface magnetic properties and electron energy levels would be required to validate.

Further investigation of the magneto-catalytic effect would benefit from a multifaceted approach across kinetic analysis and characterisation of the catalyst surface's magnetic properties to validate and build upon magneto-catalysis theoretical models. Linking kinetics to magneto-catalytic phenomena would require verification of the assumption that the pre-exponential factor is analogous to the number of active sites and the apparent activation energy is analogous to the energy barrier of adsorption / surface spin conversion. Applying a higher complexity kinetic model such as Langmuir kinetics would be useful to decouple adsorption from the apparent first order activation energy to assess if the net magnetisation of the Ionex resulted in a reduced energy barrier of spin conversion or adsorption. Assessing the use of the pre-exponential factor to describe the number of active sites would require characterisation of the surface magnetic properties to understand how the magnetic inhomogeneity changes with applied magnetic field. Magnetic characterisation may be simpler using pure crystalline ferric oxide catalysts coupled with density functional theory modelling to understand how the surface magnetic inhomogeneity changes with field strength and temperature to relate it to kinetic results. The catalyst surface magnetic properties could then be integrated into theoretical P→O conversion models with the kinetic results being useful to validate the magnetic acceleration predicted by the theoretical model. Validating magneto-catalytic theoretical models with kinetics and magnetic property characterisation could help identify potential catalysts with a high magneto-catalytic effect for use in compact cooling of superconducting motors.

6. Conclusion

Magnetic acceleration of the P→O conversion rate has been demonstrated in literature over a variety of catalysts and presents a potential use for compact catalytically cooled superconducting motors. The sensitivity of the magneto-catalytic effect to catalyst chemistry and operating temperature required experiments on conventional ferric oxide catalyst at cryogenic temperatures to assess its practical utility while also contributing a new set of data to support the development of P→O conversion theory. Applying a 2 kOe external magnetic field to Ionex indicated a 10 % P→O conversion rate deceleration at T = 90 K switching to a conversion acceleration up to 46 % at T = 40 K. Magnetic deceleration switched to acceleration at temperatures below 75 K on antiferromagnetic Ionex and is in line with the results for antiferromagnetic Cr_2O_3 in literature, where a magnetic acceleration/deceleration inversion was present near the Néel temperature. Uncertainty in the extent of magnetic acceleration significantly increased at lower temperatures with a potential 55 % relative uncertainty in the magnetic acceleration estimated at 40 K. The large uncertainty range was primarily caused by the large p-H_2 fraction measurement uncertainty relative to the change in the p-H_2 fraction across the converter in addition limitations of the first order kinetic model and pressure fluctuation. Further experiments with higher precision p-H_2 fraction measurements are critical to build assurance in the extent of magnetic acceleration possible for practical use in cooling superconducting motors.

The magnetic acceleration indicated at T = 40 K suggests that the magneto-catalytic effect could help maximise the cooling provided to a superconducting motor heat exchanger using vaporised LH_2 as a coolant. Inducing P→O cooling could increase the delivered cooling duty by 36 % for added redundancy while only requiring a volume of approximately 20 L of catalyst for a 2 MW motor if operated isothermally. Leveraging the magnetic field of the motor could reduce the required catalyst volume by 46 % compared to the non-magnetic baseline scenario and could represent the difference between meeting or exceeding the mass and volume budget of the motor cooling system. Coupling the developed kinetic model with a discretised heat exchanger model would be required to assess the impact of an axial temperature gradient on the required volume of P→O catalyst for a more detailed technical assessment of its use for cooling superconducting motors.

First order kinetic analysis suggested that the magnetic acceleration on Ionex was caused by a reduction in the number of active sites and a reduced energy barrier of adsorption/surface spin conversion, though these may be apparent changes as the model was not mechanistic. Magnetisation measurements indicated that the presence of an applied magnetic field maintained a net magnetisation of the Ionex compared to the baseline scenario, though the impact of this on the catalyst surface magnetic inhomogeneity remains unclear. Characterising the surface magnetic properties of the catalyst would be valuable to link the kinetic observations with theory to validate magneto-catalytic models proposed in literature. While a basic assessment of the magneto-catalytic effect on ferric oxide has been performed, further investigation of the magneto-catalytic effect presents value to refine P→O conversion theory while also building confidence in the use of the effect for compact catalytically cooled superconducting motors.

7. Acknowledgements

L. T. kindly acknowledges the support of the Australian Fulbright commission through a Fulbright Future Scholarship. This research was supported by an Australian Government Research Training Program Scholarship. Special thanks to Justin Jessop for designing the magnet array and Mojtaba Gohari Bahabadi, Dr. Liangguang Tang and Dr. Marcus Giansiracusa for the magnetic characterisation of Ionex.

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